Entity Tracking in Language Models

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Abstract

Keeping track of how states of entities change as a text or dialog unfolds is a key prerequisite to discourse understanding. Yet, there have been few systematic investigations into the ability of large language models (LLMs) to track discourse entities. In this work, we present a task probing to what extent a language model can infer the final state of an entity given an English description of the initial state and a series of state-changing operations. We use this task to first investigate whether Flan-T5, GPT-3 and GPT-3.5 can track the state of entities, and find that only GPT-3.5 models, which have been pretrained on large amounts of code, exhibit this ability. We then investigate whether smaller models pretrained primarily on text can learn to track entities, through finetuning T5 on several training/evaluation splits. While performance degrades for more complex splits, we find that even when evaluated on a different set of entities from training or longer operation sequences, a finetuned model can perform non-trivial entity tracking. Taken together, these results suggest that language models can learn to track entities but pretraining on text corpora alone does not make this capacity surface.

1 Introduction

A key prerequisite to long-context understanding and generating coherent text is the ability to accurately represent entities as the discourse unfolds (Karttunen, 1976; Groenendijk and Stokhof, 1991; Heim, 2002; Nieuwland and Van Berkum, 2006; Kamp et al., 2011, *i.a.*). For example, consider the following example in the context of a recipe:

(1) Put the eggs, sugar, flour, and baking powder in a bowl and mix to form a light batter. Make sure that the final batter does not contain any lumps of flour or sugar.

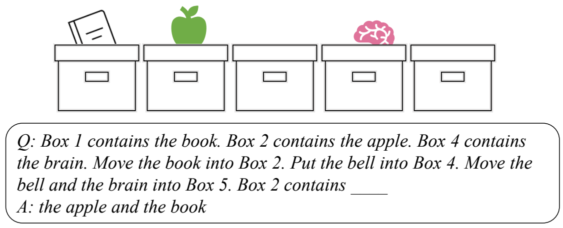

Figure 1: A sketch of our entity tracking task.

In order to understand this instruction, several distinct abilities are necessary:

New discourse entity recognition: recognizing when new discourse entities are introduced. E.g., *a bowl* introduces a new discourse entity but *the final batter* or *any lumps of ...* does not.

Coreference resolution: associating referring expressions with discourse entities. E.g., *a light batter* and *the final batter* refer to the same entity.

Discourse entity tracking: tracking the state changes made to each discourse entity. E.g., *the eggs* are put into *the bowl* and *mixed* with the other ingredients.

There exist many datasets that aim to evaluate these abilities (e.g., Walker et al., 2006; Pradhan et al., 2012; Rahman and Ng, 2012; Weston et al., 2015; Chen et al., 2018; Bamman et al., 2020; Uryupina et al., 2020) and many NLP models that aim to solve these tasks (e.g., Haghighi and Klein, 2010; Lee et al., 2011; Hill et al., 2016; Henaff et al., 2017; Ji et al., 2017; Lee et al., 2017; Bosselut et al., 2018; Gupta and Durrett, 2019a,b; Aina et al., 2019; Toshniwal et al., 2020; Wu et al., 2020). In the context of large language models (LLMs), Tenney et al. (2019), Clark et al. (2019), and Sorodoc et al. (2020) found that representations of LSTMs and Transformer-based models such as BERT (Devlin et al., 2019) do capture coreference relations. Loáiciga et al. (2022) and Schuster and Linzen (2022) found that pretrained models are able to

*Equal contribution. Author order was determined by a fair process of letting Cookie the cat pick a colored ball.

detect whether noun phrases introduce discourse entities, albeit not fully systematically.

The question of whether LLMs *can track the state of discourse entities*, however, has mostly been indirectly evaluated. Toshniwal et al. (2022) showed that GPT-2 (Radford et al., 2019) can learn to predict valid chess moves based on a compact, nonlinguistic description of previous moves. Similarly, Li et al. (2023) showed that a GPT model trained on Othello can predict valid next moves, and that these predictions are tied to the model's internal representations of the board states. Still, these results do not tell us whether LLMs track state changes expressed in natural language discourses. The most relevant evaluation is Li et al. (2021), where they tested whether model representations encode entity states described in naturalistic text. Using a probing classifier, they found that the states can be decoded from T5 (Raffel et al., 2020) and BART (Lewis et al., 2020) with high accuracy. However, as we show in a reanalysis of their results (Section 2), they do not provide definitive evidence for entity tracking. Hence, whether LLMs can track entities during the processing of natural language discourse remains an open question.

Contributions This work attempts to answer this question by developing a task targeted towards evaluating a language model's ability to track state changes of discourse entities (illustrated in Figure 1). We use this novel task to evaluate GPT-3 (Brown et al., 2020), GPT-3.5, and Flan-T5 (Chung et al., 2022) without any finetuning. We find that only models in the GPT 3.5 series, which have been trained on both text and code, are able to perform non-trivial entity tracking. We then show that a smaller language model (T5) can learn to perform non-trivial entity tracking and also demonstrates some capacity to generalize to state descriptions with more operations or with low lexical overlap. Our results suggest that language models can learn to track entities but pretraining on text corpora alone does not make this capacity surface. More broadly, our task can also serve as a useful tool for investigations into emergent world models in LMs (e.g., Li et al., 2023; Tsai et al., 2023).[1]

2 Reanalysis of Li et al. (2021)

We start from examining Li et al. (2021), the most relevant work to ours. They adapted two existing datasets, Alchemy (Long et al., 2016) and TextWorld (Côté et al., 2019), to test a model's ability to track state changes of an entity. The input to the model is a text description of the initial world state followed by state-changing instructions. Based on this description, the model is expected to identify the correct final state of each entity. For example, for Alchemy, the model receives formulaic descriptions of 7 beakers containing different amounts of colored liquids, followed by instructions that manipulate their contents such as pouring the liquid from one beaker into another, or draining a beaker. Given an input like (2), the model is expected to recognize that the first beaker has 4 units of brown liquid, the second beaker has 2 units of red liquid, and the third beaker is empty.

(2) *The first beaker has 1 green, the second beaker has 2 red, the third beaker has 3 red. Pour the last red beaker into beaker 1. Mix.*

Using such descriptions, Li et al. (2021) found that a probing classifier that takes as input the encoding of these descriptions from T5 or BART is able to correctly predict the state of 75–76% of the entities, suggesting some degree of success on entity tracking.

However, this conclusion becomes questionable when the datasets and the results are scrutinized further. Specifically, we conducted a fine-grained analysis of the success cases of the Alchemy experiment. In this experiment, the state of each beaker was probed after each state-changing instruction. Because each instruction targets at most two beakers (e.g., *pour X into Y*) and there are 7 beakers in total, there is a sparse representation of cases probing a beaker that actually underwent a change in the dataset. Indeed, 62.7% of all beaker states probed were identical to the initial state, meaning that a simple baseline that always predicts the initial state already achieves 62.7% accuracy (this is also noted by Li et al.). A second potential for shortcuts was the high rate of empty final states (32.4%).[2] For these cases, the initial state can often be entirely disregarded, due to the presence of an emptying instruction such as *Drain the fourth beaker*. This instruction alone is sufficient to predict the fourth beaker's final state independent of its initial state. Therefore, such examples are also

[1] Code and data are available at `https://github.com/sebschu/entity-tracking-lms`.

[2] This percentage also includes cases where a beaker was already empty in the initial world state description. The percentage of cases where a beaker was empty but contained some liquid in the initial description is 24.9%.

not best suited to fully assess entity tracking. Given the high prevalence of these two trivial scenarios (87.6% in total), only 12.4% of the datapoints can be considered as truly assessing state changes unfolding over a discourse context. If the accuracy is computed on the trivial and non-trivial cases separately, the probing classifier achieves 86.8% accuracy on trivial cases but only 3.1% accuracy on non-trivial cases, showing that most of the reported success derives from the trivial cases.

In summary, our reanalysis suggests that the results of Li et al. (2021) do not provide conclusive evidence for non-trivial state tracking abilities in language models.[3] However, it remains unclear whether this is due to issues with the setup or a true lack of entity tracking capacity. To this end, we propose a new behavioral evaluation.

3 Task Design and Dataset

3.1 Desiderata

The ability to track entities should be largely *independent of specific linguistic forms*. For a model that can properly track entities, it should not matter whether one talks about beakers or recipes or which specific syntactic constructions are used. This makes it an interesting ability to evaluate in the context of assessing whether and how meaning is represented, since at least classic language models are only trained on forms (Bender and Koller, 2020). At the same time, this independence of form and entity states that should hold for true entity tracking poses a challenge for evaluation, since one needs to ensure that the state of entities cannot be predicted from individual lexical items or phrases (such as the word *drain* in the Alchemy dataset, as discussed in Section 2). Furthermore, language models pretrained on large corpora may have learned common states of entities; for instance, that eggs often end up in a bowl. For these reasons, any task that evaluates entity tracking abilities should conform to the following four desiderata:

1. The probed states of entities should not follow similar distributional patterns to those that are likely to be present in the pretraining data (see also Linzen, 2020).

2. Individual words or phrases should not predict by themselves the state of an entity without considering the previous discourse in order.

3. If any data is used for demonstration, finetuning or training, the training and evaluation data should have little lexical overlap.

4. If any data is used for demonstration, finetuning or training, the task should not be solvable by slot-filling based on observed datapoints.

These properties cannot be guaranteed with naturalistic datasets such as recipes (Kiddon et al., 2015), science texts (Dalvi et al., 2019), or the Alchemy and TextWorld datasets, which have been previously used to evaluate entity tracking abilities. We therefore programmatically generated datasets for which these properties hold.

3.2 Dataset

We take inspiration from Winograd (1971) and Li et al. (2021) in designing our data. Our datasets consist of text descriptions of a particular state of the world followed by a sequence of changes. The worlds contain boxes that can be filled with objects. The objects can be placed inside the box, taken out of the box, or moved from one box to another. We define a world $\mathcal{W}$ as $\mathcal{W} = (O, n, m, e)$ where O is a set of objects, n is the number of boxes, m is the maximum number of objects one box can contain, and e is the expected number of objects in each box in the initial world states. For our datasets, we used $n = 7$, $m = 3$, $e = 2$, and used a set of nouns denoting items that can plausibly fit inside a box (e.g., *book*, *rock*, *brain*; $|O| = 100$), selected from a list of words with frequency greater than 27 in the British National Corpus (BNC; Leech et al. 2001).

A dataset consists of multiple distinct *scenarios*. A scenario consists of an initial state and a set of operations applied to this initial state. We fixed the number of operations (NumOps) in each scenario to 12. We randomly sampled 2200 scenarios, where the initial state and the 12 operations were both randomly sampled. The sampling process is designed such that only valid operations given the current world state can be sampled. The initial state and the operations were converted into naturalistic descriptions using predefined templates.

Relation to Desiderata We selected the task of moving objects across boxes because this is a domain where lexical contents of the entities do not offer cues to predict the outcome of state changes (Desideratum 1). We did not include an operation that empties a box that allows the previous

[3] Li et al. (2021) also presented two other sets of experiments. See Appendix A for details on how the other experiments exhibit similar issues.